

Weak Convergence: Selected Proofs

MBZUAI: SDS8102 – Advanced Mathematical Statistics

Portmanteau Theorem, Continuous Mapping Theorem,
Uniqueness of Characteristic Functions, and Lévy's Continuity Theorem

Throughout, X_n and X are real-valued random variables with distribution functions F_n and F , respectively.

1. Portmanteau Theorem

Theorem 1.1 (Portmanteau). *The following are equivalent:*

(i) $X_n \xrightarrow{d} X$.

(ii) For every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)].$$

(iii) For every closed $C \subseteq \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(X_n \in C) \leq \mathbb{P}(X \in C).$$

(iv) For every open $G \subseteq \mathbb{R}$,

$$\liminf_{n \rightarrow \infty} \mathbb{P}(X_n \in G) \geq \mathbb{P}(X \in G).$$

Proof. We prove

$$(i) \implies (ii) \implies (iii) \implies (iv) \implies (i).$$

$(i) \implies (ii)$. First suppose that f is continuous with compact support. Choose continuity points $a < b$ of F such that

$$\text{supp}(f) \subset (a, b).$$

For $\varepsilon > 0$, uniform continuity gives a sufficiently fine partition

$$a = x_0 < x_1 < \cdots < x_m = b,$$

whose endpoints are continuity points of F , such that the oscillation of f on each

$$I_j = (x_{j-1}, x_j]$$

is at most ε .

Choose $t_j \in I_j$ and set

$$s(x) = \sum_{j=1}^m f(t_j) \mathbf{1}_{I_j}(x).$$

Then

$$\sup_x |f(x) - s(x)| \leq \varepsilon.$$

Moreover,

$$\mathbb{P}(X_n \in I_j) = F_n(x_j) - F_n(x_{j-1}) \longrightarrow F(x_j) - F(x_{j-1}) = \mathbb{P}(X \in I_j).$$

Hence

$$\mathbb{E}[s(X_n)] \rightarrow \mathbb{E}[s(X)],$$

and therefore

$$\limsup_{n \rightarrow \infty} |\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \leq 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the claim for compactly supported f .

Now let f be bounded and continuous, with $|f| \leq M$. Choose K such that $\pm K$ are continuity points of F and

$$\mathbb{P}(|X| > K) < \varepsilon.$$

Then

$$\mathbb{P}(|X_n| > K) \rightarrow \mathbb{P}(|X| > K).$$

Let ψ_K be continuous and compactly supported, with

$$0 \leq \psi_K \leq 1, \quad \psi_K = 1 \quad \text{on } [-K, K].$$

Since $f\psi_K$ has compact support,

$$\mathbb{E}[f(X_n)\psi_K(X_n)] \rightarrow \mathbb{E}[f(X)\psi_K(X)].$$

The remaining terms are bounded by

$$M\mathbb{P}(|X_n| > K) \quad \text{and} \quad M\mathbb{P}(|X| > K).$$

Letting $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ proves

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)].$$

(ii) $\Rightarrow$ (iii). Let C be closed and define

$$f_k(x) = \max\{1 - kd(x, C), 0\}, \quad d(x, C) = \inf_{y \in C} |x - y|.$$

Then f_k is bounded and continuous, and

$$\mathbf{1}_C \leq f_k, \quad f_k \downarrow \mathbf{1}_C.$$

Thus

$$\limsup_n \mathbb{P}(X_n \in C) \leq \lim_n \mathbb{E}[f_k(X_n)] = \mathbb{E}[f_k(X)].$$

Letting $k \rightarrow \infty$ and using dominated convergence,

$$\limsup_n \mathbb{P}(X_n \in C) \leq \mathbb{P}(X \in C).$$

(iii) $\Rightarrow$ (iv). If G is open, then G^c is closed. Hence

$$\begin{aligned} \liminf_n \mathbb{P}(X_n \in G) &= 1 - \limsup_n \mathbb{P}(X_n \in G^c) \\ &\geq 1 - \mathbb{P}(X \in G^c) = \mathbb{P}(X \in G). \end{aligned}$$

(iv) $\Rightarrow$ (i). Taking complements first recovers the closed-set inequality in (iii). Let x be a continuity point of F . Since $(-\infty, x]$ is closed,

$$\limsup_n F_n(x) \leq F(x).$$

Since $(-\infty, x)$ is open,

$$\liminf_n F_n(x) \geq \mathbb{P}(X < x) = F(x^-).$$

At a continuity point, $F(x^-) = F(x)$, so

$$F_n(x) \rightarrow F(x).$$

Thus $X_n \xrightarrow{d} X$. ■

2. Continuous Mapping Theorem

Theorem 2.1 (Continuous Mapping Theorem). *Suppose*

$$X_n \xrightarrow{d} X.$$

If $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$g(X_n) \xrightarrow{d} g(X).$$

More generally, the conclusion holds if

$$\mathbb{P}(X \in D_g) = 0,$$

where D_g denotes the set of discontinuity points of g .

Proof. First suppose g is continuous. For every bounded continuous f , the composition $f \circ g$ is bounded and continuous. By Portmanteau,

$$\mathbb{E}[f(g(X_n))] = \mathbb{E}[(f \circ g)(X_n)] \rightarrow \mathbb{E}[(f \circ g)(X)] = \mathbb{E}[f(g(X))].$$

Another application of Portmanteau gives

$$g(X_n) \xrightarrow{d} g(X).$$

For the general statement, let C be closed. Then

$$\overline{g^{-1}(C)} \subseteq g^{-1}(C) \cup D_g.$$

By Portmanteau,

$$\begin{aligned} \limsup_n \mathbb{P}(g(X_n) \in C) &= \limsup_n \mathbb{P}(X_n \in g^{-1}(C)) \\ &\leq \mathbb{P}(X \in \overline{g^{-1}(C)}) \\ &\leq \mathbb{P}(g(X) \in C) + \mathbb{P}(X \in D_g) \\ &= \mathbb{P}(g(X) \in C). \end{aligned}$$

Portmanteau again yields

$$g(X_n) \xrightarrow{d} g(X). \quad \blacksquare$$

3. Moment Generating Functions and Uniqueness

Definition 3.1. The moment generating function (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

for those $t \in \mathbb{R}$ for which the expectation is finite.

Unlike characteristic functions, an MGF need not exist away from $t = 0$. However, when it exists on a neighborhood of zero, it uniquely determines the distribution.

Theorem 3.2 (Uniqueness of Moment Generating Functions). *Let X and Y be random variables. Suppose there exists $\delta > 0$ such that*

$$M_X(t) < \infty, \quad M_Y(t) < \infty, \quad |t| < \delta.$$

If

$$M_X(t) = M_Y(t) \quad \text{for every } |t| < \delta,$$

then

$$X \stackrel{d}{=} Y.$$

Proof. Because the MGFs are finite on $(-\delta, \delta)$, they extend to complex arguments

$$z = s + iu, \quad |s| < \delta,$$

through

$$M_X(z) = \mathbb{E}[e^{zX}].$$

Indeed,

$$|e^{zX}| = e^{sX},$$

and the required integrability follows from the existence of the MGF in a neighborhood of zero.

The functions

$$z \mapsto M_X(z), \quad z \mapsto M_Y(z)$$

are analytic on the strip

$$\{z \in \mathbb{C} : |\operatorname{Re} z| < \delta\}.$$

Since they agree on the real interval $(-\delta, \delta)$, the identity theorem for analytic functions implies

$$M_X(z) = M_Y(z)$$

throughout the strip.

In particular, setting $z = iu$ gives

$$\varphi_X(u) = M_X(iu) = M_Y(iu) = \varphi_Y(u) \quad \text{for every } u \in \mathbb{R}.$$

By uniqueness of characteristic functions,

$$X \stackrel{d}{=} Y.$$

■

Important point. It is not enough that all moments exist, or even that

$$\mathbb{E}[X^k] = \mathbb{E}[Y^k] \quad \text{for every } k \geq 0.$$

Different distributions can have the same sequence of moments. Existence of the MGF on an open neighborhood of 0 is a stronger condition that guarantees uniqueness.

MGF versus characteristic function. The two transforms are

$$M_X(t) = \mathbb{E}[e^{tX}], \quad \varphi_X(t) = \mathbb{E}[e^{itX}].$$

The essential distinction is

$$M_X(t) \text{ may fail to exist for } t \neq 0, \\ \varphi_X(t) \text{ exists for every } t \in \mathbb{R}.$$

When the MGF exists near zero, it determines the distribution. The characteristic function requires no such integrability assumption and always determines the distribution.

3.1. Do Moments Uniquely Determine a Distribution?

The MGF uniqueness theorem may suggest that knowing all moments

$$\mathbb{E}[X^k], \quad k = 0, 1, 2, \dots,$$

should determine the distribution of X . This is false in general.

A classical counterexample is provided by the lognormal distribution.

Example 3.3 (The lognormal moment problem). *Let*

$$X \sim \text{Lognormal}(0, 1),$$

with density

$$f(x) = \frac{1}{x\sqrt{2\pi}} \exp\left\{-\frac{(\log x)^2}{2}\right\}, \quad x > 0.$$

For $|\varepsilon| \leq 1$, define

$$f_\varepsilon(x) = f(x) [1 + \varepsilon \sin(2\pi \log x)], \quad x > 0.$$

Since

$$1 + \varepsilon \sin(2\pi \log x) \geq 0,$$

these define probability densities. For $\varepsilon \neq 0$, they are different from the original lognormal density.

We claim, however, that all of these distributions have exactly the same moments.

For any integer $k \geq 0$,

$$\int_0^\infty x^k f_\varepsilon(x) dx = \int_0^\infty x^k f(x) dx + \varepsilon \int_0^\infty x^k f(x) \sin(2\pi \log x) dx.$$

It therefore suffices to show that the second term vanishes.

Using the substitution

$$y = \log x,$$

we obtain

$$\int_0^\infty x^k f(x) \sin(2\pi \log x) dx = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{ky - y^2/2} \sin(2\pi y) dy.$$

Completing the square,

$$ky - \frac{y^2}{2} = -\frac{(y-k)^2}{2} + \frac{k^2}{2}.$$

Hence, if $Z \sim N(0, 1)$,

$$\begin{aligned} \int_0^\infty x^k f(x) \sin(2\pi \log x) dx &= e^{k^2/2} \mathbb{E}[\sin(2\pi(Z+k))] \\ &= e^{k^2/2} \mathbb{E}[\sin(2\pi Z)] \\ &= 0, \end{aligned}$$

where we used that k is an integer and symmetry of Z .

Therefore

$$\int_0^\infty x^k f_\varepsilon(x) dx = e^{k^2/2}, \quad k = 0, 1, 2, \dots,$$

independently of ε .

Thus infinitely many distinct distributions have exactly the same sequence of moments.

Why does this not contradict MGF uniqueness? For a lognormal random variable,

$$\mathbb{E}[e^{tX}] = \infty \quad \text{for every } t > 0.$$

Thus its MGF is not finite on any open neighborhood of 0.

The distinction is therefore

$$\text{same moments of every order} \not\Rightarrow \text{same distribution,}$$

whereas

$$\text{same MGF on a neighborhood of 0} \Rightarrow \text{same distribution.}$$

The MGF condition is stronger because, when it exists near zero,

$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k$$

has positive radius of convergence. In the lognormal example, the moments grow too rapidly:

$$\mathbb{E}[X^k] = e^{k^2/2},$$

and the corresponding formal power series has radius of convergence zero.

4. Characteristic Functions and Uniqueness

Definition 4.1. The characteristic function of a random variable X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{\mathbb{R}} e^{itx} dF_X(x), \quad t \in \mathbb{R}.$$

Since $|e^{itX}| = 1$, the characteristic function always exists. Moreover,

$$\varphi_X(0) = 1, \quad |\varphi_X(t)| \leq 1.$$

If X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t).$$

The fundamental fact is that φ_X determines the entire distribution of X .

Theorem 4.2 (Uniqueness of Characteristic Functions). *If*

$$\varphi_X(t) = \varphi_Y(t) \quad \text{for every } t \in \mathbb{R},$$

then

$$X \stackrel{d}{=} Y.$$

Proof. The idea is to first add a small Gaussian noise. This produces variables with smooth densities, for which ordinary Fourier inversion can be applied.

Let

$$Z \sim N(0, 1)$$

be independent of X and Y . For $\sigma > 0$, define

$$X_\sigma = X + \sigma Z, \quad Y_\sigma = Y + \sigma Z.$$

By independence,

$$\varphi_{X_\sigma}(t) = \varphi_X(t)\varphi_{\sigma Z}(t) = \varphi_X(t)e^{-\sigma^2 t^2/2},$$

and similarly

$$\varphi_{Y_\sigma}(t) = \varphi_Y(t)e^{-\sigma^2 t^2/2}.$$

Since $|\varphi_X(t)| \leq 1$,

$$|\varphi_{X_\sigma}(t)| \leq e^{-\sigma^2 t^2/2},$$

so φ_{X_σ} is integrable. Fourier inversion therefore gives the density

$$p_{X_\sigma}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_X(t) e^{-\sigma^2 t^2/2} dt.$$

Likewise,

$$p_{Y_\sigma}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_Y(t) e^{-\sigma^2 t^2/2} dt.$$

If $\varphi_X = \varphi_Y$, these two expressions are identical. Hence

$$X + \sigma Z \stackrel{d}{=} Y + \sigma Z \quad \text{for every } \sigma > 0.$$

Finally,

$$\sigma Z \xrightarrow{P} 0 \quad \text{as } \sigma \downarrow 0,$$

so

$$X + \sigma Z \xrightarrow{P} X, \quad Y + \sigma Z \xrightarrow{P} Y.$$

Thus

$$X + \sigma Z \xrightarrow{d} X, \quad Y + \sigma Z \xrightarrow{d} Y.$$

Since the distributions on the left are identical for every $\sigma > 0$, their weak limits must also be identical. Therefore

$$X \stackrel{d}{=} Y.$$

■

Interpretation. The Gaussian factor

$$e^{-\sigma^2 t^2/2}$$

makes the characteristic function integrable. This allows Fourier inversion to recover the smoothed distribution. Letting the amount of Gaussian smoothing vanish then recovers the original distribution.

5. Lévy's Continuity Theorem

The uniqueness theorem says that a characteristic function identifies a distribution. Lévy's theorem goes further: convergence of characteristic functions characterizes convergence in distribution.

Theorem 5.1 (Lévy's Continuity Theorem). *Let φ_n be the characteristic function of X_n .*

If

$$X_n \xrightarrow{d} X,$$

then

$$\varphi_n(t) \rightarrow \varphi_X(t) \quad \text{for every } t \in \mathbb{R}.$$

Conversely, suppose

$$\varphi_n(t) \rightarrow \varphi(t) \quad \text{for every } t \in \mathbb{R},$$

and φ is continuous at 0. Then φ is the characteristic function of some random variable X , and

$$X_n \xrightarrow{d} X.$$

Proof. Forward direction. Fix $t \in \mathbb{R}$. The functions

$$x \mapsto \cos(tx), \quad x \mapsto \sin(tx)$$

are bounded and continuous. Hence Portmanteau gives

$$\mathbb{E}[\cos(tX_n)] \rightarrow \mathbb{E}[\cos(tX)]$$

and

$$\mathbb{E}[\sin(tX_n)] \rightarrow \mathbb{E}[\sin(tX)].$$

Therefore

$$\varphi_n(t) = \mathbb{E}[e^{itX_n}] \rightarrow \mathbb{E}[e^{itX}] = \varphi_X(t).$$

Converse direction. The main issue is to show that the distributions of X_n cannot escape to infinity. We first establish tightness.

For $T > 0$, Fubini's theorem gives

$$\begin{aligned} \frac{1}{2T} \int_{-T}^T \varphi_n(t) dt &= \mathbb{E} \left[\frac{1}{2T} \int_{-T}^T e^{itX_n} dt \right] \\ &= \mathbb{E} \left[\frac{\sin(TX_n)}{TX_n} \right], \end{aligned}$$

where the ratio is interpreted as 1 when $X_n = 0$.

Taking real parts,

$$1 - \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt = \mathbb{E} \left[1 - \frac{\sin(TX_n)}{TX_n} \right].$$

There exists a universal constant $c > 0$ such that

$$1 - \frac{\sin y}{y} \geq c \quad \text{whenever } |y| \geq 2.$$

Therefore

$$1 - \frac{\sin(TX_n)}{TX_n} \geq c \mathbf{1}\{|X_n| \geq (2T)^{-1}\},$$

and consequently

$$\mathbb{P}(|X_n| \geq (2T)^{-1}) \leq \frac{1}{c} \left[1 - \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt \right].$$

Now $|\varphi_n(t)| \leq 1$, so dominated convergence gives

$$\frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt \longrightarrow \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi(t) dt.$$

Also,

$$\varphi(0) = \lim_n \varphi_n(0) = 1.$$

Since φ is continuous at 0,

$$\frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi(t) dt \longrightarrow 1 \quad \text{as } T \downarrow 0.$$

Hence, for every $\varepsilon > 0$, we can choose T sufficiently small so that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq (2T)^{-1}) < \varepsilon.$$

After enlarging T^{-1} if necessary to handle finitely many initial terms, we obtain

$$\sup_n \mathbb{P}(|X_n| > M) < \varepsilon$$

for some $M < \infty$. Thus $\{X_n\}$ is tight.

By tightness, every subsequence $\{X_{n_j}\}$ has a further subsequence $\{X_{n_{j_k}}\}$ such that

$$X_{n_{j_k}} \xrightarrow{d} X$$

for some random variable X .

By the forward direction,

$$\varphi_{n_{j_k}}(t) \rightarrow \varphi_X(t).$$

On the other hand, by assumption,

$$\varphi_{n_{j_k}}(t) \rightarrow \varphi(t).$$

Therefore

$$\varphi_X(t) = \varphi(t) \quad \text{for every } t.$$

Thus φ is itself a characteristic function.

By uniqueness of characteristic functions, every weakly convergent subsequence must have the same limiting distribution. Therefore the whole sequence converges:

$$X_n \xrightarrow{d} X.$$

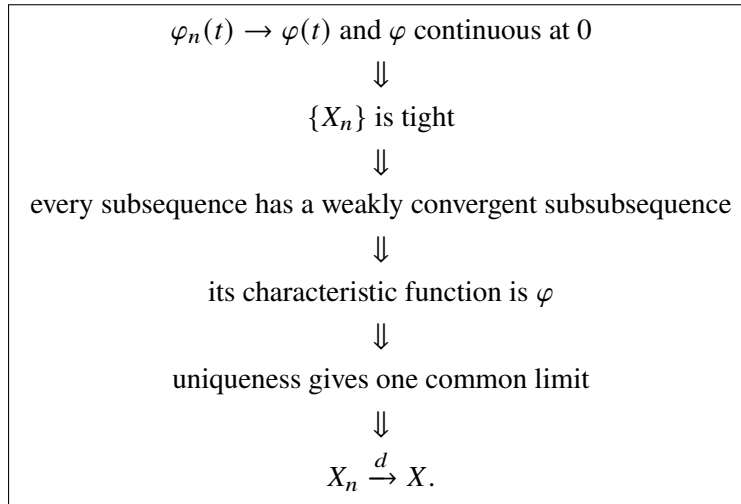
■

Why continuity at zero is needed. Pointwise convergence of characteristic functions alone does not prevent probability mass from escaping to infinity. Continuity of the limit at 0, together with

$$\varphi(0) = 1,$$

is precisely what gives the tightness needed to obtain a limiting probability distribution.

Proof structure. The converse direction can be summarized as



Summary

The main implications established above can be summarized as

$$X_n \xrightarrow{d} X \iff \mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \text{for all bounded continuous } f,$$

and

$$X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X)$$

for continuous g .

Characteristic functions provide an alternative characterization:

$$X_n \xrightarrow{d} X \implies \varphi_{X_n}(t) \rightarrow \varphi_X(t),$$

while Lévy's theorem supplies the converse when the pointwise limit is continuous at the origin.

6. The Lindeberg–Feller Central Limit Theorem

The classical central limit theorem assumes that the summands are i.i.d. The Lindeberg–Feller theorem allows the distributions of the summands to vary with n .

Consider a triangular array

$$\{X_{n,k} : 1 \leq k \leq k_n, n \geq 1\},$$

where, for each n , the random variables

$$X_{n,1}, \dots, X_{n,k_n}$$

are independent and satisfy

$$\mathbb{E}[X_{n,k}] = 0.$$

Write

$$\sigma_{n,k}^2 = \text{Var}(X_{n,k}), \quad s_n^2 = \sum_{k=1}^{k_n} \sigma_{n,k}^2.$$

Definition 6.1 (Lindeberg condition). The triangular array satisfies the Lindeberg condition if, for every $\varepsilon > 0$,

$$\frac{1}{s_n^2} \sum_{k=1}^{k_n} \mathbb{E} \left[X_{n,k}^2 \mathbf{1}_{\{|X_{n,k}| > \varepsilon s_n\}} \right] \longrightarrow 0.$$

The condition says that no substantial fraction of the total variance is contributed by exceptionally large observations.

Theorem 6.2 (Lindeberg–Feller CLT). Suppose that, within each row, $X_{n,1}, \dots, X_{n,k_n}$ are independent, centered random variables and that $s_n^2 > 0$. If the Lindeberg condition holds, then

$$\frac{1}{s_n} \sum_{k=1}^{k_n} X_{n,k} \xrightarrow{d} N(0, 1).$$

Why the Lindeberg condition is natural

Define

$$Y_{n,k} = \frac{X_{n,k}}{s_n}.$$

Then

$$\sum_{k=1}^{k_n} \mathbb{E}[Y_{n,k}^2] = 1,$$

and the Lindeberg condition becomes

$$\sum_{k=1}^{k_n} \mathbb{E} \left[Y_{n,k}^2 \mathbf{1}_{\{|Y_{n,k}| > \varepsilon\}} \right] \longrightarrow 0.$$

Thus the total variance is normalized to one, while the contribution from summands of non-negligible size vanishes.

In particular, Lindeberg implies the Feller condition

$$\max_{1 \leq k \leq k_n} \frac{\sigma_{n,k}^2}{s_n^2} \longrightarrow 0.$$

Indeed, for every $\varepsilon > 0$,

$$\mathbb{E}[Y_{n,k}^2] \leq \varepsilon^2 + \mathbb{E} \left[Y_{n,k}^2 \mathbf{1}_{\{|Y_{n,k}| > \varepsilon\}} \right].$$

Taking the maximum over k gives

$$\max_k \mathbb{E}[Y_{n,k}^2] \leq \varepsilon^2 + \sum_k \mathbb{E} \left[Y_{n,k}^2 \mathbf{1}_{\{|Y_{n,k}| > \varepsilon\}} \right],$$

and then $n \rightarrow \infty$ followed by $\varepsilon \downarrow 0$ proves the claim.

Proof idea via characteristic functions

Let

$$S_n = \sum_{k=1}^{k_n} Y_{n,k}.$$

By independence,

$$\varphi_{S_n}(t) = \prod_{k=1}^{k_n} \varphi_{n,k}(t),$$

where

$$\varphi_{n,k}(t) = \mathbb{E}[e^{itY_{n,k}}].$$

For small x ,

$$e^{itx} = 1 + itx - \frac{t^2 x^2}{2} + r_t(x),$$

where

$$r_t(x) = o(x^2) \quad (x \rightarrow 0).$$

Since $\mathbb{E}[Y_{n,k}] = 0$,

$$\varphi_{n,k}(t) = 1 - \frac{t^2}{2} \mathbb{E}[Y_{n,k}^2] + \mathbb{E}[r_t(Y_{n,k})].$$

The Lindeberg condition allows the remainder terms to be summed uniformly:

$$\sum_{k=1}^{k_n} \mathbb{E}[r_t(Y_{n,k})] \rightarrow 0.$$

Therefore

$$\varphi_{n,k}(t) = 1 - \frac{t^2}{2} \mathbb{E}[Y_{n,k}^2] + \text{negligible remainder}.$$

Since

$$\sum_{k=1}^{k_n} \mathbb{E}[Y_{n,k}^2] = 1$$

and

$$\max_k \mathbb{E}[Y_{n,k}^2] \rightarrow 0,$$

taking logarithms of the product gives

$$\log \varphi_{S_n}(t) \rightarrow -\frac{t^2}{2}.$$

Hence

$$\varphi_{S_n}(t) \rightarrow e^{-t^2/2}.$$

The limiting function is the characteristic function of $N(0, 1)$. Lévy's continuity theorem therefore gives

$$S_n \xrightarrow{d} N(0, 1).$$

Remark. The Lindeberg condition is sufficient for asymptotic normality. Under the additional Feller condition

$$\max_k \frac{\sigma_{n,k}^2}{s_n^2} \rightarrow 0,$$

it is also necessary. Thus, for infinitesimal triangular arrays, Lindeberg's condition precisely characterizes when the normalized sum has a Gaussian limit.

Application: Asymptotic Normality of Weighted Estimators

A major advantage of the Lindeberg–Feller theorem is that the summands need not be identically distributed. This makes it useful for studying estimators that can be written as weighted sums.

Suppose

$$\widehat{\theta}_n - \theta = \sum_{i=1}^n w_{n,i} \varepsilon_i,$$

where the ε_i are independent with

$$\mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma_i^2.$$

Then

$$s_n^2 = \text{Var}(\widehat{\theta}_n - \theta) = \sum_{i=1}^n w_{n,i}^2 \sigma_i^2.$$

If, for every $\varepsilon > 0$,

$$\frac{1}{s_n^2} \sum_{i=1}^n \mathbb{E} \left[w_{n,i}^2 \varepsilon_i^2 \mathbf{1}_{\{|w_{n,i} \varepsilon_i| > \varepsilon s_n\}} \right] \rightarrow 0,$$

then the Lindeberg–Feller theorem gives

$$\frac{\widehat{\theta}_n - \theta}{s_n} \xrightarrow{d} N(0, 1).$$

Thus an asymptotically valid confidence interval is

$$\widehat{\theta}_n \pm z_{1-\alpha/2} s_n,$$

or the corresponding interval obtained after replacing s_n by a consistent estimator.

Example: heteroskedastic regression. Consider

$$Y_i = x_i^\top \beta + \varepsilon_i, \quad \mathbb{E}[\varepsilon_i] = 0, \quad \text{Var}(\varepsilon_i) = \sigma_i^2,$$

with independent but not necessarily identically distributed errors. For a fixed vector a ,

$$a^\top (\widehat{\beta}_{\text{OLS}} - \beta) = \sum_{i=1}^n a^\top (X^\top X)^{-1} x_i \varepsilon_i.$$

This is a triangular-array weighted sum. Under a suitable Lindeberg condition preventing any single observation from dominating, Lindeberg–Feller yields asymptotic normality of the OLS estimator.

This is one reason triangular-array CLTs appear naturally in asymptotic statistics: observations may contribute different amounts of information and may have different variances.

7. The Law of the Iterated Logarithm

The law of large numbers and the central limit theorem describe two different aspects of the partial sums

$$S_n = \sum_{i=1}^n (X_i - \mu).$$

If $X_1, X_2, \dots$ are i.i.d. with

$$\mathbb{E}[X_1] = \mu, \quad \text{Var}(X_1) = \sigma^2 < \infty,$$

then the strong law gives

$$\frac{S_n}{n} \xrightarrow{a.s.} 0,$$

while the central limit theorem gives

$$\frac{S_n}{\sigma\sqrt{n}} \xrightarrow{d} N(0, 1).$$

Neither result describes the largest fluctuations of S_n along a single infinite sample path. This is the role of the law of the iterated logarithm.

Theorem 7.1 (Law of the Iterated Logarithm). *Let $X_1, X_2, \dots$ be i.i.d. with*

$$\mathbb{E}[X_1] = \mu, \quad 0 < \text{Var}(X_1) = \sigma^2 < \infty.$$

Let

$$S_n = \sum_{i=1}^n (X_i - \mu).$$

Then

$$\limsup_{n \rightarrow \infty} \frac{S_n}{\sigma\sqrt{2n \log \log n}} = 1 \quad a.s.,$$

and

$$\liminf_{n \rightarrow \infty} \frac{S_n}{\sigma\sqrt{2n \log \log n}} = -1 \quad a.s.$$

Equivalently, along almost every sample path,

$$S_n$$

reaches arbitrarily close to both envelopes

$$\pm \sigma\sqrt{2n \log \log n}$$

infinitely often, while eventually remaining within any slightly larger envelope.

More precisely, for every $\varepsilon > 0$,

$$|S_n| \leq (1 + \varepsilon)\sigma\sqrt{2n \log \log n}$$

eventually almost surely, while

$$S_n \geq (1 - \varepsilon)\sigma\sqrt{2n \log \log n}$$

infinitely often almost surely, and similarly for the negative boundary.

CLT versus LIL

At first sight, the CLT and LIL may appear to give different scales. The CLT says

$$S_n = O_P(\sqrt{n}),$$

whereas the LIL shows that

$$|S_n|$$

can be as large as order

$$\sqrt{n \log \log n}$$

infinitely often.

There is no contradiction.

The CLT concerns the distribution of S_n at a *single large time* n . For fixed M ,

$$\mathbb{P}\left(\frac{|S_n|}{\sigma\sqrt{n}} > M\right)$$

is approximately the Gaussian tail probability

$$\mathbb{P}(|Z| > M).$$

The LIL instead considers an *entire infinite sample path* and asks how large the fluctuations become infinitely often. Even very rare fluctuations eventually occur when infinitely many times are examined.

Thus the three classical results describe different scales:

SLLN	:	$S_n = o(n)$	a.s.,
CLT	:	$S_n/\sqrt{n}$ has a nondegenerate distributional limit,	
LIL	:	$\limsup \frac{S_n}{\sqrt{2\sigma^2 n \log \log n}} = 1$	a.s.

Statistical perspective. The CLT and LIL answer fundamentally different inferential questions:

CLT:	How uncertain is an estimator at one large sample size?
LIL:	How large can its fluctuations become when monitored repeatedly over time?

Thus the CLT underlies fixed-sample inference, while the LIL provides intuition for sequential inference and time-uniform guarantees.